APPM 4600 — Midterm 2 Review Sheet

Here is a list of important concepts to review from the material for Midterm 1.

- **Fixed point iteration for nonlinear systems of equations.**
  - What is required for convergence?
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- **Systems of Nonlinear equations and root finding methods (nD):**
  (a) Consider the fixed point iteration applied to $F : R^n \to R^n$. How would you apply this to a root finding problem? Under what conditions on its Jacobian G is F a contractive map?
  (b) Derive Newton's method for systems of nonlinear equations. Under what assumptions does this method converge quadratically? How does the Newton's method for nonlinear systems of equations differ from Newton's method for scalar equations?

- **Quasi-Newton methods:**
  (a) What are the different quasi-Newton methods we talked about?
  (b) What is the motivation for creating these techniques?
  (c) How do these techniques address the difficulty that motivated their development? Can you, for instance, compare the cost per step with Newton?

- **Interpolation:**
  (a) What are the different interpolation techniques?
  (b) What data is needed for each one?
  (c) What is the form for the representation? How is it constructed? How stable is it?
  (d) What is the error for each technique? Given fixed nodes, how would you create an upper bound on the error? Given a function that you can evaluate easily, what can you control to minimize the error?

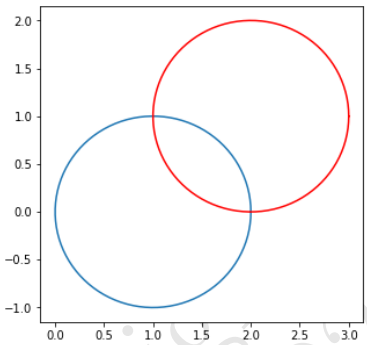
- **Approximation:**
  (a) What is the difference between discrete approximation, L^2 approximation and interpolation?
  (b) How do we construct each approximation technique?
  (c) When do you use each approximation technique?
  (d) What does it mean for polynomials to be orthogonal with respect to a weight function?
  (e) Is a trig approximation (Fourier) an L^2 approximation?
  (f) For what functions is a trig approximation useful?

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(g) What is a Padé approximation? Why does it behave better than the Taylor polynomial?

Note: Below are some practice problems. This list of problems is not exhaustive. You should not expect the exam to look exactly like this. These are simply some problems that you should be able to solve using the material from class. Making sure you understand all the concepts and algorithms introduced in class, homeworks and flipped days is the best way to prepare you for the exam. Be sure to understand why, when and how an algorithm works in addition to the limitations.

1. Consider the problem of finding the intersections between two circles of radius 1, with centers at $(1, 0)$ and $(2, 1)$, respectively.



From this picture, we can clearly see two roots: $(1, 1)$ and $(2, 0)$. We can write this as a system of two nonlinear equations for coordinates (x, y):

$$F_1(x) = (x-1)^2 + y^2 - 1 = 0$$
$$F_2(x) = (x-2)^2 + (y-1)^2 - 1 = 0$$

Where $F(x) = [F_1(x) \quad F_2(x)]$. We find the Jacobian of $F(x)$:

$$J_F(x) = \begin{bmatrix} 2(x-1) & 2y \\ 2(x-2) & 2(y-1) \end{bmatrix}$$

(a) We use the Newton method starting at initial point $(x0, y0) = (1, 0)$. We immediately get a complaint from our software that reads 'LinAlgError: Singular Matrix'. Compute the Jacobian at this initial point and explain why we got this message.

(b) After running the Newton method for a good amount of initial points $(x0, y0)$, we notice a pattern. Iterations starting above the line $y = x - 1$ converge to $(1, 1)$ and iterations starting below it converge to $(2, 0)$. If we start exactly at this line, we get the same error message we got in (a). Plug in $y = x - 1$ in the formula for the Jacobian and explain why that might be the case.

2. Consider the Broyden Quasi-Newton method to find roots for a system of $n = 1000$ equations of the form $F(x) = 0$. Given $B_0 = J_F(x_0)$ (Jacobian at initial guess).

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- Write down the first step of the Broyden method.
- Assume you have access to the matrix B_0^{-1}, or that given vectors v, you can compute the product $B_0^{-1}v$. Compare the following two pseudocodes to compute the update B_1, and determine which one is preferable (and why):
  (a) Compute the update $B_1 = B_0 + \frac{r_n^T \Delta x_n}{\Delta x_n^T \Delta x_n}$. Invert or use a linear solve to compute the next Broyden step.
  (b) Compute the update for the inverse $B_1^{-1} = B_0^{-1} + u_n v_n^T$ (using Sherman-Morrison formula), use it to compute the next Broyden step.

3. Consider the method of steepest descent for root finding of the system $F(x) = 0$. We use it to minimize the function:

$$g(x) = ||F(x)||^2 = \sum_{j=1}^{n} F_j(x)^2$$

   (a) Derive an expression for the gradient of $g(x)$ involving $F(x)$ and its derivatives (Jacobian matrix $J_F(x)$).
   (b) The method of steepest descent takes steps of the form:

$$x_{n+1} = x_n - \alpha_n \nabla g(x_n)$$

   Say you decide to pick $\alpha_n = 1$ for all n. Explain: What can go wrong with this "lazy steepest descent"?

4. Let $f(x) = \sqrt{1+x}$ with interpolation nodes $x_0 = 0$, $x_1 = 1/2$ and $x_2 = 1$.
   (a) Create the highest order approximation possible using these interpolation nodes.
   (b) Derive a bound on the error when approximation $f(x)$ for $x \in [0, 1]$ using the polynomial from part (a).

5. Using Lagrange interpolation as your inspiration, create a polynomial of degree at most 2 such that $p(x_0) = y_0$, $p'(x_1) = y_2$ and $p''(x_1) = y_3$.

6. What does it mean for a cubic spline to be natural, clamped or periodic? How do you impose these conditions, and how do they change the linear system (matrix and right-hand-side) to find the coefficients of the cubic "pieces" of these splines?

7. Show that the Hermite polynomial interpolant H_{2n+1} is the unique polynomial of degree $\leq 2n+1$ which enforces $H_{2n+1}(x_j) = f(x_j)$ and $H'_{2n+1}(x_j) = f'(x_j)$ for $j = 0, \ldots, n$.

8. Using the error bound for linear interpolation, derive an error bound for piecewise linear interpolation of $f(x)$ on $[a, b]$.

9. Consider the task of fitting a quadratic polynomial to the data in the table using least squares method

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i	x_i	y_i
1	0	1.0000
2	0.25	1.2480
3	0.50	1.6487
4	0.75	2.1170
5	1.00	2.7183

(a) Write the general form for the quadratic.

(b) Derive a square linear system for the identifying the coefficients of the polynomial. (There are two ways of getting this system.)

10. Find the linear least squares polynomial approximation on the interval [-1, 1] with respect to the weight function $w(x) = 1$ of $f(x) = e^x$.

11. Find a [1/1] Padé approximant (numerator of degree $m = 1$, denominator of degree $n = 1$) for $f(x) = e^x$ around $x = 0$.

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